

MTH2021 Mathematical Methods 2

Self-study sheet 2

1. Show that each function below is non-analytic by identifying two independent directions, along which the rates of change of the function are different:

$$(a) \quad f(z) = \operatorname{Im} z \qquad (b) \quad f(z) = \frac{z^2}{z^{*2}}$$

2. Use the Cauchy-Riemann equations to determine whether, and where, the following functions of z are analytic:

$$\begin{array}{lll} (a) \quad z^4 & (b) \quad z^2 - (z^*)^2 & (c) \quad \operatorname{Im} z \\ (d) \quad 1/(z + i) & (e) \quad \cos z & (f) \quad \sinh z \end{array}$$

3. Differentiate $f(z) = z^n$, for n a positive integer, from first principles.
4. Using the definitions of the circular and hyperbolic functions, show that

$$\begin{array}{ll} (a) \quad \cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2 \\ (b) \quad \cos(iz) = \cosh z \\ (c) \quad \sinh(z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2 \\ (d) \quad \sinh(iz) = i \sin z \end{array}$$

Find $\sinh(i\pi)$ and $\cosh(i\pi/3)$.

5. Find all values of z such that

$$(a) \quad z = \log(-2) \qquad (b) \quad \exp(iz) = 1 + i$$
